FLEET TEAM DIGEST

737MAX-FTD-29-17001

Issue Title : 737 Hydraulic Reservoir Pressurization Gauge Design Improvement

Airplane Model	ATA	Minor Model(s)
737MAX	2909-03	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
03/14/2017	08/01/2017	07/25/2017

Estimated Completion Date	Next Update Date

Revision Description

Updated to provide closing actions for this project.

Applicability

All 737MAX airplanes.

Description

In recent years, Boeing has received numerous operator reports of reservoir pressurization gauge failures in the 737 fleet, usually resulting in leakage and/or an inoperative gauge. Several failed gauges have been examined by Boeing and we have determined the typical root cause of failures to be slow-growth fatigue fracturing of the gauge's internal bourdon tube.
Furthermore, Boeing has determined that this type of failure can be attributed to vibration and that the failure generally manifests itself via either the inability of the hydraulic system to maintain head pressure on the reservoir or the gauge becoming inoperative and failing to indicate the actual head pressure.

Background

Boeing has examined numerous failed gauges and has determined the typical root cause of failures to be slow-growth fatigue fracturing of the gauge's bourdon tube. Furthermore, Boeing has determined that this type of failure can be attributed to vibration. After reviewing the original qualification testing requirements of the pressure gauge, it was determined that vibration testing was performed to vibration levels lower than is currently being experienced in service. The current production gauge has since been tested to the updated vibration standard and it has been found that the pressure gauge fails short of the current service life requirements. As such, the above findings have prompted Boeing to pursue a more robust gauge design that can adequately meet or exceed the aircraft's service life requirements.

Status

Boeing and Perma-Cal, the current production gauge supplier, have developed and approved a more robust reservoir pressurization gauge that aligns with 737 wheel well vibration requirements.

A prototype gauge with a new "triple wound" bourdon tube was tested against Boeing wheel well vibration requirements and proved to be a significant improvement over the former production gauge (P/N 90-900-1032).

Boeing has reviewed and approved the supplier engineering drawings and process specifications.
Furthermore, Boeing has updated the 737 engineering drawings and committed the change on production aircraft Line Number (L/N) 6500, delivered in late July 2017.

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Major milestones for the process are provided in the MILESTONES section, below, recognizing that they may require adjustment as the work progresses.

Interim Action

None at this time.

Final Action

Boeing has incorporated the improved "triple-wound" bourdon tube gauge (P/N 90-900-1052) in 737 production on aircraft L/N 6500.

Furthermore, Boeing Service Letter 737-SL-29-134 (dated 27-JUL-2017) recommends that operators remove and replace in-service gauges on an attrition basis.

NOTE: The updated P/N 90-900-1052 reservoir pressure gauge is fully interchangeable with the prior gauge options (P/N 90-900-1032 and P/N G3721) on the 737MAX.

The 737MAX Aircraft Illustrated Parts Catalog (AIPC) reflects this interchangeability in the 15-JUL-2017 revision.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: 19-DEC-2012 (Complete)

Solution Selected: 07-SEP-2016 (Complete)

Change Committed: 16-JAN-2017 (Complete)

Production Incorporation: 27-JUL-2017 (Complete)

Service Letter Available: 27-JUL-2017 (Complete)

Parts Available: 03-JUL-2017 (Complete)

Operator Action

As stated above, Boeing has approved and incorporated the improved P/N 90-900-1052 reservoir pressurization gauge in 737NG/MAX production in order to mitigate in-service failures.

Operators experiencing failures of P/N 90-900-1032 hydraulic reservoir pressure gauges may wish to replace these gauges on an attrition basis with improved P/N 90-900-1052 gauges.

Replacement can be carried out using the AMM 29-09-03/401.

Improved P/N 90-900-1052 hydraulic reservoir pressure gauges became available via Boeing Spares in July 2017.

NOTE: Boeing would like to inform operators that if a hydraulic reservoir pressurization gauge has been found failed or leaking, the aircraft can fly with a blank installed in place of the reservoir pressurization gauge per MMEL 29-09-02 for up to 10 calendar days.

Part Information

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The reservoir pressurization gauge (see attached illustration) is located in the main landing gear wheel well and provides on-ground indication of the System A and B reservoir head pressure via the reservoir balance line.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Letter (SL)	737-SL-29-134		07/27/2017
Fleet Team Economic (FTE)	FTE-29-11009		
Illustrated Parts Catalog (IPC)	29-09-03-01 Item 10		
Fleet Team Economic (FTE)	FTE-29-14013		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Reservoir Pressurization Gauge (737MAX approved spare, prior production gaug)		Perma-Cal	90-900-1032
Reservoir Pressurization Gauge (737MAX approved spare)		Rogerson	G3721
Reservoir Pressurization Gauge (current production gauge)		Perma-Cal	90-900-1052

Related Categories

- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board
- In Service Occurrences : null
- New Equipment : Information on new parts available now or soon to be introduced

Attachments:

a) Reservoir Pressurization Gauge

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